

RESEARCH

Open Access



Quantum secret sharing for edge computing security using locally distinguishable eight-qudit entangled states

Feng Liu¹, Chunliu Cai^{2*}, Xianmin Wang^{3*}, Zhongliang Ma³ and Sib0 Huang⁴

*Correspondence:
202010092@gdufs.edu.cn;
xianmin@gzhu.edu.cn

¹ Scientific Research Department,
Guangzhou Preschool Teachers
College, Guangzhou, China

² Library, Guangdong University
of Foreign Studies, Guangzhou,
China

³ Institution of Artificial
Intelligence, Guangzhou
University, Guangzhou, China

⁴ Melbourne Business School,
University of Melbourne,
Parkville, Australia

Abstract

Maintaining confidentiality and reliability in distributed quantum environments remains a significant challenge. To this end, we propose a quantum secret sharing (QSS) scheme based on the local distinguishability of orthogonal eight-qudit entangled states. Compared to the previous seven-particle model, this scheme significantly expands the quantum state space, enhances security, and improves flexibility. We systematically construct 22 types of eight-qudit entangled states, define their corresponding judgment spaces, and introduce a new distinction rule to simplify the local discrimination process. Based on this framework, we develop a general (k, n) -threshold quantum synchronization sharing protocol and verify its effectiveness using a $(7, 8)$ -threshold example on the IBM Qiskit platform. Results show that this scheme is resilient to multiple attack types and provides a scalable approach for secure quantum information sharing in future distributed quantum systems, with potential applications in fields such as quantum-enhanced edge computing.

Keywords: Quantum secret sharing, Edge computing, Quantum computation, Orthogonal entangled states, Local distinguishability

1 Introduction

In recent years, the rapid development of quantum computing has introduced new paradigms for information processing, while also posing significant challenges to classical cryptographic systems. As we move toward an era of distributed quantum networks, ensuring secure multi-party collaboration and computation has become a critical research direction. Quantum secret sharing (QSS), one of the most promising techniques in quantum cryptography, provides a powerful framework for this challenge. It allows a secret to be distributed among multiple participants such that only a predefined threshold number of them can cooperatively recover it, while smaller subsets learn nothing. This mechanism is fundamental for building trust and enabling secure computation in decentralized environments.

A prominent example of such a distributed environment is edge computing, which processes vast amounts of data close to the source and has an increasing need for security protocols that go beyond classical limitations. The principles of quantum mechanics

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

offer inherent advantages in resisting eavesdropping and supporting higher levels of security. Since Hillery et al. first introduced QSS in 1998, numerous schemes have been proposed and experimentally verified [1–4]. These approaches make QSS particularly relevant to future distributed ecosystems, including quantum-enhanced edge networks [5–7]. However, many existing QSS schemes still face challenges such as reliance on a trusted dealer, complex joint measurements, or limited scalability, which hinder their practical deployment in dynamic and heterogeneous environments. In response, a significant line of research has focused on developing schemes based on local operations and classical communication (LOCC) to improve practicality and address these security concerns.

To overcome these limitations, Rahaman et al. [8] proposed a simple and highly effective (k, n) threshold QSS scheme based on local quantum operations and classical communication (LOCC), termed as the LOCC-QSS scheme. The core idea involves encoding a secret into a set of distinguishable orthogonal multipartite entangled states by a dealer. These particles are then distributed to agents following specific rules, allowing the reconstruction of the secret only when a sufficient number of participants collaborate. Additionally, the scheme employs eigenvalue equations to detect eavesdropping attacks, offering robust resistance against such attacks. However, Yang et al. [9] swiftly identified potential information leakage risks in the scheme proposed in reference [8]. In their work [9], they presented a standard $(2, n)$ threshold QSS scheme based on LOCC to quantify the leakage of secret information. To quantify the leakage of secret information, Yang et al. proposed two attack methods, unambiguous state discrimination attack and guessing attack. In 2017, Wang et al. [10] introduced the concept of judgment space, outlining conditions for quantum states to resist unambiguous state discrimination attacks based on judgment space properties. Nonetheless, their expression of judgment space was considered not entirely perfect. In a separate effort, Liu et al. [11] proposed a QSS scheme based on seven-particle orthogonal entangled states, but this scheme exhibited certain security risks and lacked flexibility and broad applicability in practical use. Additionally, Xu et al. [12] proposed a multi-party quantum secret sharing scheme based on GHZ state entanglement swapping and measurement. In their approach, participants independently generated classical bits, enhancing both the transmission efficiency and security of the scheme. In 2023, Li et al. [13] proposed a (t, n) threshold quantum secret sharing scheme that ensures fairness while leveraging the privacy attributes of secure multi-party computation, enabling the repeated use of participants' secret shares. In 2023, Kuo et al. [14] identified vulnerabilities in traditional multi-party quantum secret sharing (QSS) protocols based on quantum secure direct communication, specifically issues related to step-skipping and unequal participant privileges. To address these issues, the authors proposed a novel multi-party QSS scheme that ensures independent and equal authority for each participant. Security analysis demonstrated that the proposed scheme can resist attacks from malicious participants while preserving the advantages of quantum direct communication and reducing communication overhead. In 2024, Qin et al. [15] proposed a practical quantum secret sharing (QSS) scheme that achieves high efficiency, security, and flexibility simultaneously, addressing key limitations in multiuser QSS applications. The protocol utilizes a continuous-variable eight-partite bound entangled state and requires only two nondegenerated optical

parametric amplifiers, showcasing an efficient use of quantum hardware. In 2024, Tian et al. [16] proposed a novel dynamic multi-party to multi-party quantum secret sharing (QSS) protocol based on Bell states, enabling secure distribution of secrets among dynamically changing user groups. The protocol employs Bell states with simplified local unitary operations, reducing implementation complexity and enhancing scalability in practical quantum communication systems. In 2024, Mawlia et al. [17] proposed a threshold quantum secret sharing (QSS) scheme that integrates mutual identity authentication using mutually unbiased bases, enhancing the security of quantum communication. In this protocol, Alice distributes the secret shares via a symmetric bivariate polynomial and a trusted participant shares a GHZ state with a qualified group for secret reconstruction. By employing quantum Fourier transform (QFT) and unitary operations tied to individual secret shares, the scheme ensures that unrevealed secrets are never transmitted. In 2025, Bai et al. [18] developed a new verifiable quantum secret sharing scheme by constructing a special multi-particle entanglement masking process based on orthogonal arrays. In this scheme, an unknown quantum state is embedded into a multi-particle entangled state for distribution and the secret quantum state is reconstructed through carefully designed measurement and unitary operations. As revealed by the above analysis, the scheme proposed by Liu et al. [11] fails to resolve the issues of limited availability of quantum states in the design and the inconvenience in selecting states suitable for local distinguishability protocols.

Although previous research has made progress in LOCC-QSS, discrimination spaces, and multi-party threshold schemes, several challenges remain: (i) the available sets of orthogonal entangled states are limited and inconvenient to select, which affects the flexibility and scalability of protocols; (ii) the formulation and usage rules of discrimination spaces are not sufficiently intuitive and unified, making it difficult to support systematic distinguishability analysis under LOCC; (iii) there is a lack of a construction framework that balances universal (k, n) threshold design with rigorous proofs of distinguishability thresholds. To address these issues, this paper constructs 22 classes of eight-dimensional, eight-particle orthogonal entangled states to expand the quantum state design space and enhance protocol flexibility. Furthermore, it proposes novel discrimination rules for judgment spaces, characterizing distinguishability through aggregation patterns of participants' local measurement results. Finally, a universal (k, n) threshold QSS protocol is designed, with implementation and verification demonstrated using a $(7, 8)$ example. The security analysis shows that the proposed scheme can resist intercept-resend attacks and entangle-measure attacks.

The main contributions of this work are summarized as follows:

- Novel eight-qudit entangled states construction: We systematically construct 22 distinct classes of orthogonal eight-qudit entangled states, expanding the quantum state space significantly compared to the prior seven-particle model [15]. This expansion provides a richer foundation for encoding and sharing secrets.
- Refined judgment space and distinction rules: We introduce a novel and generalized distinction rule within the judgment space framework. This rule offers a more precise and intuitive characterization for the local distinguishability of multipartite entangled states, addressing the limitations in the expression of judgment space.

- General (k, n) threshold QSS protocol: Based on the newly proposed distinction rule, we develop a general (k, n) threshold quantum secret sharing protocol. The protocol's security is rooted in the local distinguishability of the constructed states, ensuring that no fewer than k participants can recover the secret.
- Practical verification and security analysis: We provide a concrete instantiation of a $(7, 8)$ threshold scheme and validate its feasibility through simulations on the IBM Qiskit platform. Furthermore, a comprehensive security analysis demonstrates the scheme's robustness against various attacks, including intercept-resend and entangled-measurement attacks.

The paper is structured as follows: Sects. 2.1 and 2.2 introduce preliminary knowledge and the construction of 22 classes of eight-particle entangled states, along with new discrimination rules and theoretical results; Sect. 2.3 presents the universal (k, n) threshold scheme and a $(7, 8)$ case study; Sect. 3 details the simulation setup, results, and security verification; Sect. 4 discusses advantages, limitations, and potential applications; and Sect. 5 concludes the paper.

2 Methods

2.1 Preliminaries

To ensure compliance with theoretical principles in the construction of quantum states and the design of the proposed scheme, the following section introduces several definitions and concepts employed throughout this paper.

(1) d -dimensional Hilbert space n -qubit orthogonal entangled states:

Let $\{|0\rangle, |1\rangle, \dots, |d-1\rangle\}$ be a set of orthonormal bases in a d -dimensional Hilbert space H . An n -qudit quantum system can be denoted as $H^{\otimes n}$. Let $H^{\otimes k}$ and $H^{\otimes(n-k)}$ be two associated subsystems of $H^{\otimes n}$. $|e_i^{(k)}\rangle$ and $|e_j^{(n-k)}\rangle$ are their respective orthonormal bases. Let $|\varphi_1\rangle$ and $|\varphi_2\rangle$ be a pair of n -qudit entangled states in the Hilbert space $H^{\otimes n}$, which can be represented in an appropriate basis as:

$$\begin{cases} |\varphi_1\rangle = \sum_{i,j} a_{ij} |e_i^{(k)}\rangle |e_j^{(n-k)}\rangle \\ |\varphi_2\rangle = \sum_{i,j} a'_{ij} |e_i^{(k)}\rangle |e_j^{(n-k)}\rangle \end{cases} \quad (1)$$

where for any i, j , $\sum_{i,j} |a_{ij}|^2 = 1 = \sum_{i,j} |a'_{ij}|^2$, and $a_{ij}a'_{ij} = 0$.

(2) Judgment space:

If $S_1^{(k)}$ and $S_2^{(k)}$ are subspaces of the Hilbert space $H^{\otimes n}$, their representations are as follows:

$$\begin{cases} S_1^{(k)} = \left\{ |e_i^{(k)}\rangle \mid \exists j, \text{ s.t. } a_{ij} \neq 0 \right\} \\ S_2^{(k)} = \left\{ |e_i^{(k)}\rangle \mid \exists j, \text{ s.t. } a'_{ij} \neq 0 \right\} \end{cases} \quad (2)$$

where $S_1^{(k)}$ and $S_2^{(k)}$ are respectively referred to as the k -level judgment spaces of $|\varphi_1\rangle$ and $|\varphi_2\rangle$. Correspondingly, the $(k-1)$ -level judgment spaces of $|\varphi_1\rangle$ and $|\varphi_2\rangle$ are denoted as $S_1^{(k-1)}$ and $S_2^{(k-1)}$.

(3) Generalized Pauli operators:

The representation of generalized Pauli operators in a d -dimensional Hilbert space is as follows:

$$\begin{cases} X = \sum_{j=0}^{d-1} |j+1\rangle\langle j| \\ Z = \sum_{j=0}^{d-1} \omega^j |j\rangle\langle j| \end{cases} \quad (3)$$

where $\omega = e^{\frac{2\pi i}{d}}$.

Throughout this paper, we fix the local dimension at $d = 8$. Accordingly, qudits refer to 8-dimensional systems, and all generalized Pauli operators and measurement bases are defined over $\mathbb{Z}_8$.

2.2 Construction of eight-qudit entangled states and novel rules for distinguishing judgment spaces

2.2.1 Construction of eight-qudit entangled states

Let $|\varphi\rangle = \sum_{i_1, i_2, i_3, i_4, i_5, i_6, i_7, i_8} |i_1 i_2 i_3 i_4 i_5 i_6 i_7 i_8\rangle$ be an eight-qudit entangled state containing eight particles. Based on the different states of the particles in $|\varphi\rangle$, it can be categorized into eight classes under permutation, as detailed below:

- (1) The scenario where eight particles have one state: $|aaaaaaaa\rangle (a = 0, \dots, 7)$
- (2) The scenario where eight particles have two different states: $|aaaaaaab\rangle, |aaaaaabb\rangle, |aaaaabbb\rangle, |aaaabbbb\rangle (a, b = 0, \dots, 7, a > b)$
- (3) The scenario where eight particles have three different states: $|aaaaaabc\rangle, |aaaaabbc\rangle, |aaaabbbc\rangle, |aaabbbcc\rangle, |aabbccdd\rangle (a, b, c = 0, \dots, 7, a > b > c)$
- (4) The scenario where eight particles have four different states: $|aaaaabcd\rangle, |aaaabbcd\rangle, |aaabbbcd\rangle, |aabbccdd\rangle (a, b, c, d = 0, \dots, 7, a > b > c > d)$
- (5) The scenario where eight particles have five different states: $|aaaabcde\rangle, |aaabbcde\rangle, |aabbccde\rangle (a, b, c, d, e = 0, \dots, 7, a > b > c > d > e)$
- (6) The scenario where eight particles have six different states: $|aaabcdef\rangle, |aabbccdef\rangle (a, b, c, d, e, f = 0, \dots, 7, a > b > c > d > e > f)$
- (7) The scenario where eight particles have seven different states: $|aabcdefg\rangle (a, b, c, d, e, f, g = 0, \dots, 7, a > b > c > d > e > f > g)$
- (8) The scenario where eight particles have eight different states: $|abcdefgh\rangle (a, b, c, d, e, f, g, h = 0, \dots, 7, a > b > c > d > e > f > g > h)$

Based on the above analysis, a total of 22 different cases were obtained. To simplify the calculation of normalization coefficients and the subsequent description of the judgment space, we constrain the values of a, b, c, d, e, f , and g, h . Let $a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3, e = a + 4, f = a + 5, g = a + 6, h = a + 7$, where the "+" indicates modulo 8 addition.

The 22 obtained eight-particle entangled states are as follows:

$$|\varphi_0\rangle = \frac{1}{\sqrt{8}} \sum_{a=0}^7 |aaaaaaaa\rangle \quad (4)$$

$$|\varphi_1\rangle = \frac{1}{\sqrt{64}} \sum_{P_i \in P} \sum_{a=0; b=j+1}^7 P_i(|aaaaaaab\rangle) \quad (5)$$

$$|\varphi_2\rangle = \frac{1}{\sqrt{224}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaaaabb\rangle) \quad (6)$$

$$|\varphi_3\rangle = \frac{1}{\sqrt{448}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaaabbb\rangle) \quad (7)$$

$$|\varphi_4\rangle = \frac{1}{\sqrt{560}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaabbbb\rangle) \quad (8)$$

$$|\varphi_5\rangle = \frac{1}{\sqrt{448}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaaaaabc\rangle) \quad (9)$$

$$|\varphi_6\rangle = \frac{1}{\sqrt{1344}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaaaabbc\rangle) \quad (10)$$

$$|\varphi_7\rangle = \frac{1}{\sqrt{2240}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaaabbbc\rangle) \quad (11)$$

$$|\varphi_8\rangle = \frac{1}{\sqrt{3360}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaaabbcc\rangle) \quad (12)$$

$$|\varphi_9\rangle = \frac{1}{\sqrt{4480}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaabbbcc\rangle) \quad (13)$$

$$|\varphi_{10}\rangle = \frac{1}{\sqrt{2688}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaaaabcd\rangle) \quad (14)$$

$$|\varphi_{11}\rangle = \frac{1}{\sqrt{6720}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaaabbcd\rangle) \quad (15)$$

$$|\varphi_{12}\rangle = \frac{1}{\sqrt{8960}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaabbbcd\rangle) \quad (16)$$

$$|\varphi_{13}\rangle = \frac{1}{\sqrt{13440}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaabbccd\rangle) \quad (17)$$

$$|\varphi_{14}\rangle = \frac{1}{\sqrt{20160}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aabbccdd\rangle) \quad (18)$$

$$|\varphi_{15}\rangle = \frac{1}{\sqrt{13440}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4}^7 P_i(|aaaabcde\rangle) \quad (19)$$

$$|\varphi_{16}\rangle = \frac{1}{\sqrt{26880}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4}^7 P_i(|aaabbcde\rangle) \quad (20)$$

$$|\varphi_{17}\rangle = \frac{1}{\sqrt{20160}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4}^7 P_i(|aabbccde\rangle) \quad (21)$$

$$|\varphi_{18}\rangle = \frac{1}{\sqrt{53760}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4; f=a+5}^7 P_i(|aaabcdef\rangle) \quad (22)$$

$$|\varphi_{19}\rangle = \frac{1}{\sqrt{80640}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4; f=a+5}^7 P_i(|aabbccdef\rangle) \quad (23)$$

$$|\varphi_{20}\rangle = \frac{1}{\sqrt{161280}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4; f=a+5; g=a+6}^7 P_i(|aabcdefg\rangle) \quad (24)$$

$$|\varphi_{21}\rangle = \frac{1}{\sqrt{40320}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3; e=a+4; f=a+5; g=a+6; h=a+7}^7 P_i(|abcdefgh\rangle) \quad (25)$$

where P_i is the set of all possible distinct permutations.

2.2.2 Normalization of the eight-qudit entangled states

The normalization factors presented in Eqs. (4)–(25) are rigorously derived from the general structure of the eight-qudit entangled states. This section provides a detailed mathematical derivation to justify these normalization constants.

All 22 classes of states share a common structural form. A general state $|\phi_s\rangle$ in our scheme can be expressed as:

$$|\phi_s\rangle = \frac{1}{\sqrt{N}} \sum_{a=0}^{d-1} \omega_s^{f_s(a)} |r_1(a), r_2(a), \dots, r_8(a)\rangle, \quad (26)$$

where:

- $d = 8$ is the dimension of each qudit
- $\omega = e^{2\pi i/d} = e^{2\pi i/8}$ is the primitive d th root of unity
- $f_s : \mathbb{Z}_d \rightarrow \mathbb{Z}_d$ is a phase function specific to the state class s
- $r_j(a)$ for $j = 1, \dots, 8$ is a permutation function that determines the state of the j th qudit given the index a
- N is the normalization constant ensuring $\langle \phi_s | \phi_s \rangle = 1$

To prove normalization, we compute the inner product:

$$\langle \phi_s | \phi_s \rangle = \left(\frac{1}{\sqrt{N}} \sum_{a=0}^{d-1} \omega^{-f_s(a)} |r_1(a), \dots, r_8(a)\rangle \right) \left(\frac{1}{\sqrt{N}} \sum_{b=0}^{d-1} \omega^{f_s(b)} |r_1(b), \dots, r_8(b)\rangle \right) \quad (27)$$

$$= \frac{1}{N} \sum_{a=0}^{d-1} \sum_{b=0}^{d-1} \omega^{f_s(b)-f_s(a)} \prod_{j=1}^8 \langle r_j(a) | r_j(b) \rangle \quad (28)$$

The computational basis states are orthonormal:

$$\langle r_j(a) | r_j(b) \rangle = \delta_{r_j(a), r_j(b)} \quad (29)$$

Since each r_j is a permutation over the set $\mathbb{Z}_d$, it is a bijective mapping. Therefore, the condition $r_j(a) = r_j(b)$ for all $j = 1, \dots, 8$ can only hold if $a = b$. This gives:

$$\prod_{j=1}^8 \delta_{r_j(a), r_j(b)} = \delta_{a,b} \quad (30)$$

Applying this to the double summation:

$$\begin{aligned} \langle \phi_s | \phi_s \rangle &= \frac{1}{N} \sum_{a=0}^{d-1} \sum_{b=0}^{d-1} \omega^{f_s(b)-f_s(a)} \delta_{a,b} \\ &= \frac{1}{N} \sum_{a=0}^{d-1} \omega^{f_s(a)-f_s(a)} = \frac{1}{N} \sum_{a=0}^{d-1} 1 = \frac{d}{N} \end{aligned} \quad (31)$$

For normalization, we require $\langle \phi_s | \phi_s \rangle = 1$, which implies $N = d = 8$ for the fundamental case. For more complex states involving permutations, N equals the total number of distinct terms in the superposition, as shown in the following examples.

For the state $|\varphi_0\rangle$ in Eq. (4):

$$|\varphi_0\rangle = \frac{1}{\sqrt{8}} \sum_{a=0}^7 |a, a, a, a, a, a, a, a\rangle \quad (32)$$

There are exactly 8 distinct terms in the superposition, hence $N = 8$.

For the state $|\varphi_1\rangle$ in Eq. (5):

$$|\varphi_1\rangle = \frac{1}{\sqrt{64}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaaaaab\rangle) \quad (33)$$

There are 8 choices for a , and for each a , there are $\binom{8}{1} = 8$ distinct permutations of the pattern $|aaaaaaab\rangle$, resulting in $8 \times 8 = 64$ distinct terms, hence $N = 64$.

This counting principle applies uniformly to all 22 classes of states, with the normalization factor N in each case equaling the total number of distinct permutations of the respective pattern under the specified constraints, ensuring proper normalization and orthogonality of the constructed eight-qudit entangled states.

2.2.3 Construction of novel rules for distinguishing judgment spaces

Before presenting the novel discrimination rules, we first define the notation for the judgment spaces of the constructed eight-particle entangled states. Let $S_i^{(k)}$ denote the k -level judgment space of the entangled state $|\varphi_i\rangle$, where $i = 0, 1, \dots, 21$ corresponds to the 22 states defined in Sect. 2.2.1, and k denotes the level of the judgment space.

According to the definition of judgment space provided in the Preliminaries (Sect. 2.1), we can compute the k -level judgment spaces of $|\varphi_i\rangle$ for $i = 0, 1, \dots, 21$. As an example, we list the 7-level judgment spaces below:

(1) When $k = 7$, the 7-level judgment spaces of $|\varphi_i\rangle$ for $i = 0, 1, \dots, 21$ are as follows:

$$S_0^{(7)} = \{|aaaaaaa\rangle; a = 0, \dots, 7\}$$

$$S_1^{(7)} = \{P_i(|jjjjjj\rangle); P_i(|aaaaaab\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1\}$$

$$S_2^{(7)} = \{P_i(|aaaaabb\rangle); P_i(|aaaaaab\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1\}$$

$$S_3^{(7)} = \{P_i(|aaaabbb\rangle); P_i(|aaaaabb\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1\}$$

$$S_4^{(7)} = \{P_i(|aaabbbb\rangle); P_i(|aaaabbb\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1\}$$

$$S_5^{(7)} = \{P_i(|aaaaabc\rangle); P_i(|aaaaaab\rangle); P_i(|aaaaaac\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2\}$$

$$S_6^{(7)} = \{P_i(|aaaaaac\rangle); P_i(|aaaaabc\rangle); P_i(|aaaaabb\rangle) \\ P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2\}$$

$$S_7^{(7)} = \{P_i(|aaabbbc|); P_i(|aaaabbc|); P_i(|aaaabbb|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2\}$$

$$S_8^{(7)} = \{P_i(|aaabbcc|); P_i(|aaaabcc|); P_i(|aaaabbc|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2\}$$

$$S_9^{(7)} = \{P_i(|aabbbcc|); P_i(|aaabbcc|); P_i(|aaabbbc|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2\}$$

$$S_{10}^{(7)} = \{P_i(|aaaabcd|); P_i(|aaaaacd|); P_i(|aaaaabd|); P_i(|aaaaabc|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3\}$$

$$S_{11}^{(7)} = \{P_i(|aaabbcd|); P_i(|aaaabcd|); P_i(|aaaabbc|); P_i(|aaaabbd|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3\}$$

$$S_{12}^{(7)} = \{P_i(|aabbbcd|); P_i(|aaabbcd|); P_i(|aaabbbd|); P_i(|aaabbbc|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3\}$$

$$S_{13}^{(7)} = \{P_i(|aabbccd|); P_i(|aaabccd|); P_i(|aaabbcd|); P_i(|aaabbbc|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3\}$$

$$S_{14}^{(7)} = \{P_i(|abbccdd|); P_i(|aabccdd|); P_i(|aabbccd|); P_i(|aabbccd|)\}$$

$$P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3\}$$

$$S_{15}^{(7)} = \{P_i(|aaabcde|); P_i(|aaaacde|); P_i(|aaaabde|); P_i(|aaaabce|);$$

$$P_i(|aaaabcd|)\} P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3,$$

$$e = a + 4\}$$

$$S_{16}^{(7)} = \{P_i(|aabbcd|); P_i(|aaabcd|); P_i(|aaabbd|); P_i(|aaabbc|);$$

$$P_i(|aaabbbcd|)\} P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3,$$

$$e = a + 4\}$$

$$S_{17}^{(7)} = \{P_i(|abbccde|); P_i(|aabccde|); P_i(|aabbcd|); P_i(|aabbce|);$$

$$P_i(|aabbccdd|)\} P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, d = a + 3,$$

$$e = a + 4\}$$

$$S_{18}^{(7)} = \{P_i(|aabcdef\rangle); P_i(|aaacdef\rangle); P_i(|aaabdef\rangle); P_i(|aaabcef\rangle); \\ P_i(|aaabcdf\rangle); P_i(|aaabcde\rangle) \mid P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, \\ d = a + 3, e = a + 4, f = a + 5\}$$

$$S_{19}^{(7)} = \{P_i(|abbcddef\rangle); P_i(|aabcdef\rangle); P_i(|aabbdef\rangle); P_i(|aabbcef\rangle); \\ P_i(|aabbcdf\rangle); P_i(|aabbcdde\rangle) \mid P_i \in P, a = 0, \dots, 7, b = a + 1, c = a + 2, \\ d = a + 3, e = a + 4, f = a + 5\}$$

$$S_{20}^{(7)} = \{P_i(|abcdefg\rangle); P_i(|aacdefg\rangle); P_i(|aabdefg\rangle); P_i(|aabcefg\rangle); \\ P_i(|aabcdfg\rangle); P_i(|aabcdeg\rangle); P_i(|aabcdef\rangle) \mid P_i \in P, a = 0, \dots, 7, b = a + 1, \\ c = a + 2, d = a + 3, e = a + 4, f = a + 5\}$$

$$S_{21}^{(7)} = \{|abcdefg\rangle; a \neq b \neq c \neq d \neq e \neq f \neq g; a, b, c, d, e, f, g = 0, \dots, 7\}$$

To facilitate the discrimination of quantum states employed in constructing local distinction schemes, this section proposes a novel discrimination rule to characterize the possible measurement outcomes when seven participants perform operations on seven particles. The formal definition of the rule is as follows:

$$(m * xA, n * yA)^k$$

where $n, m, x, y = 0, \dots, 7$; $k = 1, \dots, 7$. Here, m and n represent the number of groups with identical measurement outcomes, x and y represent the counts of identical measurement outcomes and k denotes the level. For example, $(7A)^7$ denotes seven identical measurement results. $(5A, 2A)^7$ denotes a group of five identical results and a group of two identical results. $(3A, 2 * 2A)^7$ denotes a group of three identical results and two groups of two identical results. $(3A, 2A)^7$ denotes a group of three identical results and a group of two identical results. $(0A)^7$ denotes seven different measurement results.

In this paper, $X_i^{(k)}$ ($i = 0, 1, \dots, 21$) denotes the k -level judgment space of $|\varphi_i\rangle$. The distinguishing rules for the 7-level judgment spaces of $|\varphi_i\rangle$ ($i = 0, 1, \dots, 21$) are represented as follows:

$$\begin{aligned}
X_0^{(7)} &= \{(7A)^7\}, & X_1^{(7)} &= \{(7A)^7, (6A)^7\} \\
X_2^{(7)} &= \{(5A, 2A)^7, (6A)^7\}, & X_3^{(7)} &= \{(4A, 3A)^7, (5A, 2A)^7\} \\
X_4^{(7)} &= \{(4A, 3A)^7\}, & X_5^{(7)} &= \{(5A)^7, (6A)^7\} \\
X_6^{(7)} &= \{(4A, 2A)^7, (5A)^7, (5A, 2A)^7\} & X_7^{(7)} &= \{(3A, 3A)^7, (4A, 2A)^7, (4A, 3A)^7\} \\
X_8^{(7)} &= \{(3A, 2 * 2A)^7, (4A, 2A)^7\} & X_9^{(7)} &= \{(3A, 2 * 2A)^7, (3A, 3A)^7\} \\
X_{10}^{(7)} &= \{(4A)^7, (5A)^7\} & X_{11}^{(7)} &= \{(3A, 2A)^7, (4A)^7, (4A, 2A)^7\} \\
X_{12}^{(7)} &= \{(3A, 2A)^7, (3A, 3A)^7\} & X_{13}^{(7)} &= \{(3 * 2A)^7, (3A, 2A)^7, (3A, 2 * 2A)^7\} \\
X_{14}^{(7)} &= \{(3 * 2A)^7\} & X_{15}^{(7)} &= \{(3A)^7, (4A)^7\} \\
X_{16}^{(7)} &= \{(2A, 2A)^7, (3A)^7, (3A, 2A)^7\} & X_{17}^{(7)} &= \{(2A, 2A)^7, (3 * 2A)^7\} \\
X_{18}^{(7)} &= \{(2A)^7, (3A)^7\} & X_{19}^{(7)} &= \{(2A)^7, (2A, 2A)^7\} \\
X_{20}^{(7)} &= \{(0A)^7, (2A)^7\} & X_{21}^{(7)} &= \{(0A)^7\}
\end{aligned}$$

When $k = 6$, similarly, $(6A)^6$ represents six identical measurement results. $(4A, 2A)^6$ represents one group of four identical measurements and another group of two identical measurements. $(3 * 2A)^6$ represents three groups of two identical measurements. $(0A)^6$ represents six different measurement results.

$$\begin{aligned}
X_0^{(6)} &= \{(6A)^6\}, & X_1^{(6)} &= \{(6A)^6, (5A)^6\} \\
X_2^{(6)} &= \{(4A, 2A)^6, (5A)^6, (6A)^6\} & X_3^{(6)} &= \{(3A, 3A)^6, (4A, 2A)^6, (5A)^6\} \\
X_4^{(6)} &= \{(3A, 3A)^6, (4A, 2A)^6\} & X_5^{(6)} &= \{(4A)^6, (5A)^6, (6A)^6\} \\
X_6^{(6)} &= \{(3A, 2A)^6, (4A)^6, (4A, 2A)^6, (5A)^6\} & & \\
X_7^{(6)} &= \{(3A, 2A)^6, (3A, 3A)^6, (4A)^6, (4A, 2A)^6\} & X_9^{(6)} &= \{(3A, 2A)^6, (3 * 2A)^6, (3A, 3A)^6\} \\
X_8^{(6)} &= \{(3 * 2A)^6, (3A, 2A)^6, (4A, 2A)^6, (4A)^6\} & & \\
X_{10}^{(6)} &= \{(3A)^6, (4A)^6, (5A)^6\} & & \\
X_{11}^{(6)} &= \{(2A, 2A)^6, (3A)^6, (3A, 2A)^6, (4A)^6, (4A, 2A)^6\} & & \\
X_{12}^{(6)} &= \{(3A)^6, (2A, 2A)^6, (3A, 2A)^6, (3A, 3A)^6\} & X_{13}^{(6)} &= \{(2A, 2A)^6, (3 * 2A)^6, (3A, 2A)^6\} \\
X_{14}^{(6)} &= \{(3 * 2A)^6, (2A, 2A)^6, (3A, 2A)^6\} & X_{15}^{(6)} &= \{(2A)^6, (3A)^6, (4A)^6\} \\
X_{16}^{(6)} &= \{(2A)^6, (3A)^6, (2A, 2A)^6, (3A, 2A)^6\} & X_{17}^{(6)} &= \{(2A)^6, (2A, 2A)^6, (3 * 2A)^6\} \\
X_{18}^{(6)} &= \{(2A)^6, (3A)^6, (0A)^6\} & X_{19}^{(6)} &= \{(2A)^6, (0A)^6, (2A, 2A)^6\} \\
X_{20}^{(6)} &= \{(2A)^6, (0A)^6\} & X_{21}^{(6)} &= \{(0A)^6\}
\end{aligned}$$

To verify that the proposed scheme satisfies the conditions for local distinguishability, three theorems are provided for subsequent validation along with their proofs.

1 Theorem 1

The states $|\varphi_1\rangle$ and $|\varphi_2\rangle$ are two orthogonal n -qudit entangled states. Their k -level new judgment space distinction rules are denoted as $X_{(1)}^{(k)}$ and $X_{(2)}^{(k)}$. If the intersection between $X_{(1)}^{(k)}$ and $X_{(2)}^{(k)}$ is empty, then $|\varphi_1\rangle$ and $|\varphi_2\rangle$ can be distinguished by no fewer than k participants using LOCC.

1 Proof

If there are k participants who perform local measurements on their particles using the computational basis $\{|j\rangle\}_{j=0}^{d-1}$, randomly selecting k measurement outcomes and performing a tensor product operation between them, it results in $|e_i^{(k)}\rangle$. Because the intersection between $X_{(1)}^{(k)}$ and $X_{(2)}^{(k)}$ is empty, the new judgment space distinction rule for $|e_i^{(k)}\rangle$ belongs to either $X_{(1)}^{(k)}$ or $X_{(2)}^{(k)}$. If the new judgment space distinction rule for $|e_i^{(k)}\rangle$ belongs to $X_{(1)}^{(k)}$, then the shared state is $|\varphi_1\rangle$, otherwise the shared state is $|\varphi_2\rangle$.

1 Theorem 2

If $|\varphi_1\rangle$ and $|\varphi_2\rangle$ are two n -qudit orthogonal entangled states, and their $(k-1)$ -level judgment spaces are $X_{(1)}^{(k-1)}$ and $X_{(2)}^{(k-1)}$ respectively, if $X_{(1)}^{(k-1)}$ equals $X_{(2)}^{(k-1)}$, then $|\varphi_1\rangle$ and $|\varphi_2\rangle$ cannot be unambiguously distinguished by fewer than k participants using LOCC.

1 Proof

If k participants individually measure their particles using the computational basis $\{|j\rangle\}_{j=0}^{d-1}$, and then perform a tensor product on k randomly chosen measurement results, resulting in $|e_i^{(k-1)}\rangle$. Because $X_{(1)}^{(k-1)}$ and $X_{(2)}^{(k-1)}$ are equal, the new judgment space distinction rule for $|e_i^{(k-1)}\rangle$ belongs to both $X_{(1)}^{(k-1)}$ and $X_{(2)}^{(k-1)}$. Therefore, $|\varphi_1\rangle$ and $|\varphi_2\rangle$ cannot be unambiguously distinguished by fewer than k participants using LOCC.

Based on Theorem 1 and Theorem 2, it is possible to design a (k, n) threshold quantum secret sharing scheme that withstands unambiguous attacks. Satisfying the first condition is relatively easy, but meeting both conditions simultaneously can be challenging. Hence, the following theorem is used to weaken the second condition.

1 Theorem 3

Let $|\varphi_0\rangle$, $|\varphi_1\rangle$ and $|\varphi_2\rangle$ be a set of n -qudit orthogonal entangled states. Their $(k-1)$ level new judgment space distinction rule satisfies:

$$\begin{cases} X_0^{(k-1)} \subseteq X_1^{(k-1)} \cup X_2^{(k-1)} \\ X_1^{(k-1)} \subseteq X_0^{(k-1)} \cup X_2^{(k-1)} \\ X_2^{(k-1)} \subseteq X_0^{(k-1)} \cup X_1^{(k-1)} \end{cases} \quad (34)$$

then $|\varphi_0\rangle$, $|\varphi_1\rangle$ and $|\varphi_2\rangle$ cannot be unambiguously distinguished by fewer than k participants using LOCC.

1 Proof

If there are $k-1$ participants who perform local measurements on their particles using the computational basis $\{|j\rangle\}_{j=0}^{d-1}$, randomly select $k-1$ measurement results, and compute the tensor product among them, resulting in $|e_i^{(k-1)}\rangle$. Since new judgment space distinction rule of $|e_i^{(k-1)}\rangle$ belongs to $X_{(0)}^{(k-1)}$ and $X_{(0)}^{(k-1)}$ is in the union of $X_{(1)}^{(k-1)}$ and $X_{(2)}^{(k-1)}$, new judgment space distinction rule of $|e_i^{(k-1)}\rangle$ is at least part of two among $X_{(0)}^{(k-1)}$, $X_{(1)}^{(k-1)}$ and $X_{(2)}^{(k-1)}$. Therefore, $k-1$ participants cannot unambiguously distinguish the shared quantum state using LOCC.

2.3 A (k, n) threshold QSS scheme based on LOCC

2.3.1 Our construction

According to the analysis above, a threshold (k, n) QSS scheme based on LOCC can be proposed. Secret recovery requires the cooperation of no fewer than k participants, thus being resilient against unambiguous attacks. Assuming Alice wants to share a secret among $Bob_i (i = 1, 2, \dots, n)$ and secret recovery requires the collaboration of no fewer than k participants. The detail steps are as follows: **Preparation phase:** Alice first prepares a set of n -qudit orthogonal entangled states $\{|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_5\rangle, |\psi_6\rangle\}$. Then from this set of orthogonal entangled states, she randomly selects M ($M > n$) quantum states, each one represented as $(a_1^j, a_2^j, \dots, a_n^j)$ $j = 1, 2, \dots, m$, where $a_1^j, a_2^j, \dots, a_n^j$ represent the ordered particles of the j th entangled state. Additionally, the k -level new judgment space distinction rule of $\{|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_5\rangle, |\psi_6\rangle\}$ complies with the requirements of theorem 1, and the $(k-1)$ -level new judgment space distinction rule satisfies the requirements of theorem(3). For clarity, the selected quantum states are transformed into the following sequential format:

$$\begin{aligned} S_1 &= \{a_1^1, a_2^1, \dots, a_n^1\}, \\ S_2 &= \{a_1^2, a_2^2, \dots, a_n^2\}, \\ &\vdots \\ S_n &= \{a_1^n, a_2^n, \dots, a_n^n\} \end{aligned} \quad (35)$$

Distribution phase: To resist attacks from dishonest participants, Alice randomly prepares n different sequences, denoted as $r_i = \prod_i(1, 2, \dots, M)$, where $i \in \{1, 2, \dots, n\}$ and $\prod_i$ represents different permutations of the sequence $(1, 2, \dots, M)$. Alice then uses the order of sequence r_i to shuffle the order of sequences $S_1, S_2, \dots, S_n$ from the first step. Consequently, the sequence $S_1, S_2, \dots, S_n$ becomes $S'_1, S'_2, \dots, S'_n$. Subsequently, Alice sends the sequence $S'_k, (k = 1, 2, \dots, n)$ to each corresponding Bob_i . Alice only sends quantum bits, without transmitting any information about the sequences. Thus, only Alice has knowledge of the sequence information.

Measurement phase: Alice randomly selects certain moments $\{t_s\}_{s=1}^\mu$ ($\{t_s\}_{s=1}^\mu \subset \{1, 2, \dots, M\}$), and simultaneously computes p_i , where p_i is a permutation of $\{1, 2, \dots, u\}$. Then, Alice prepares n sequences $C_i = \{[\sigma_i(t_{p_i(s)}), \prod_i(t_{p_i(s)})]\}_{s=1}^\mu$. Subsequently, she sends C_i to Bob_i . Upon receiving the sequence C_i , Bob_i measures the $\sigma_i(t_{p_i(s)})$ basis on the $\prod_i(t_{p_i(s)})$ th qudit and forwards the measurement result $v(t_{p_i(s)})$ to Alice.

Verification phase: Alice checks whether the error rate of the measurement results meets the predetermined value. By examining this, Alice can easily detect if there is any eavesdropping. If there is any eavesdropping, she terminates the process, otherwise she continues.

Recovery phase: In the absence of eavesdropping, Alice announces the sequence order r_i to each participant. Participants then perform measurements in the correct order. Here, $|\varphi_0\rangle$ represents $a = 0$, $|\varphi_1\rangle$ represents $a = 1$, $|\varphi_2\rangle$ represents $a = 2$, $|\varphi_3\rangle$ represents $a = 3$, $|\varphi_4\rangle$ represents $a = 4$, $|\varphi_5\rangle$ represents $a = 5$, and $|\varphi_6\rangle$ represents $a = 6$. The relationship between classical bit values and orthogonal entangled states was sent by Alice to Bob_i in advance through a secure channel. With the inference that no fewer than k participants can distinguish $\{|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_5\rangle, |\psi_6\rangle\}$, the secret can be reconstructed.

2.3.2 Instantiation

The following example illustrates the scheme with a (7, 8) threshold.

Distribution phase: Alice prepares a set of 8-qubit orthogonal entangled states

$$|\varphi_0\rangle = \frac{1}{\sqrt{8}} \sum_{a=0}^7 |aaaaaaa\rangle \quad (36)$$

$$|\varphi_2\rangle = \frac{1}{\sqrt{224}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaaabb\rangle) \quad (37)$$

$$|\varphi_4\rangle = \frac{1}{\sqrt{560}} \sum_{P_i \in P} \sum_{a=0; b=a+1}^7 P_i(|aaaabbbb\rangle) \quad (38)$$

$$|\varphi_8\rangle = \frac{1}{\sqrt{3360}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2}^7 P_i(|aaaabbcc\rangle) \quad (39)$$

$$|\varphi_{10}\rangle = \frac{1}{\sqrt{2688}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaaaabcd\rangle) \quad (40)$$

$$|\varphi_{12}\rangle = \frac{1}{\sqrt{8960}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aaabbbcd\rangle) \quad (41)$$

$$|\varphi_{14}\rangle = \frac{1}{\sqrt{20160}} \sum_{P_i \in P} \sum_{a=0; b=a+1; c=a+2; d=a+3}^7 P_i(|aabbccdd\rangle) \quad (42)$$

The quantum states chosen from the previous text can be represented as $|S(a, b_t)\rangle$, where $t \in \{1, 2, \dots, M\}$, a represents the chosen quantum state and $b_t = (1_t, 2_t, \dots, n_t)$ represents the bit positions of the chosen quantum state at time t .

Distribution phase: Same as the previous phase.

Measurement phase: Alice selecting random measurement bases from the sets $\{X, Z^2, Z^3, Z^4, Z^5\}$, $\{X, Z^2, Z^3, Z^4\}$, $\{X, Z^2, Z^3\}$, $\{X, Z^4\}$, $\{X, Z^3\}$, $\{X, Z^2\}$, and X when the quantum state $|S(a, b_t)\rangle$ is $|\varphi_0\rangle, |\varphi_2\rangle, |\varphi_4\rangle, |\varphi_8\rangle, |\varphi_{10}\rangle, |\varphi_{12}\rangle$ or $|\varphi_{14}\rangle$. Here, X and Z are generalized Pauli operators on an 8-dimensional Hilbert space and the set $\{O\} = \{X^{\otimes 8}, (Z^2)^{\otimes 8}, (Z^3)^{\otimes 8}, (Z^4)^{\otimes 8}, (Z^5)^{\otimes 8}\}$.

Verification phase: $O_{t_s} \in \{O\}$, which is an n -element binary vector. Alice selects states corresponding to O_{t_s} as eigenstates, satisfying the characteristic equation:

$$O_{t_s} |S[a, r(b_{t_p(s)})]\rangle = \lambda(a, t_s) |S[a, r(b_{t_p(s)})]\rangle \quad (43)$$

If for all $O_{t_s} \in \{O\}$, $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_0\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^2)^{\otimes 8}$ or $(Z^3)^{\otimes 8}$ or $(Z^4)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_2\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^2)^{\otimes 8}$ or $(Z^3)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_4\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^5)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_8\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^4)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_{10}\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^3)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_{10}\rangle$.

If $O_{t_s} = X^{\otimes 8}$ or $(Z^2)^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_{12}\rangle$.

If $O_{t_s} = X^{\otimes 8}$ and $\lambda(a, t_s) = +1$, then $|S[a, r(b_t)]\rangle = |\varphi_{14}\rangle$.

Recovery phase: Same as the previous phase.

The following describes the scheme as a (7, 8) threshold QSS based on LOCC. The 7-level new judgment space distinction rules for $\{|\psi_0\rangle, |\psi_2\rangle, |\psi_4\rangle, |\psi_8\rangle, |\psi_{10}\rangle, |\psi_{12}\rangle, |\psi_{14}\rangle\}$ are as follows:

$$X_0^{(7)} = \{(7A)^7\}$$

$$X_2^{(7)} = \{(5A, 2A)^7, (6A)^7\}$$

$$X_4^{(7)} = \{(4A, 3A)^7\}$$

$$X_8^{(7)} = \{(3A, 2 * 2A)^7, (4A, 2A)^7\}$$

$$X_{10}^{(7)} = \{(4A)^7, (5A)^7\}$$

$$X_{12}^{(7)} = \{(3A, 2A)^7, (3A, 3A)^7\}$$

$$X_{14}^{(7)} = \{(3 * 2A)^7\}$$

Because $X_0^{(7)} \cap X_2^{(7)} \cap X_4^{(7)} \cap X_8^{(7)} \cap X_{10}^{(7)} \cap X_{12}^{(7)} \cap X_{14}^{(7)} = \emptyset$, according to Theorem 1, seven participants can collaborate to distinguish the prepared entangled states.

The 6-level new judgment space distinction rules for $\{|\psi_0\rangle, |\psi_2\rangle, |\psi_4\rangle, |\psi_8\rangle, |\psi_{10}\rangle, |\psi_{12}\rangle, |\psi_{14}\rangle\}$ are as follows:

$$X_0^{(6)} = \{(6A)^6\}$$

$$X_2^{(6)} = \{(4A, 2A)^6, (5A)^6, (6A)^6\}$$

$$X_4^{(6)} = \{(3A, 3A)^6, (4A, 2A)^6\}$$

$$X_8^{(6)} = \{(3 * 2A)^6, (3A, 2A)^6, (4A, 2A)^6, (4A)^6\}$$

$$X_{10}^{(6)} = \{(3A)^6, (4A)^6, (5A)^6\}$$

$$X_{12}^{(6)} = \{(3A)^6, (2A, 2A)^6, (3A, 2A)^6, (3A, 3A)^6\}$$

$$X_{14}^{(6)} = \{(3 * 2A)^6, (2A, 2A)^6, (3A, 2A)^6\}$$

From the above judgment space distinction rules, it can be deduced that:

$$X_0^{(6)} \subseteq X_2^{(6)} \cup X_4^{(6)} \cup X_8^{(6)} \cup X_{10}^{(6)} \cup X_{12}^{(6)} \cup X_{14}^{(6)}$$

$$X_2^{(6)} \subseteq X_0^{(6)} \cup X_4^{(6)} \cup X_8^{(6)} \cup X_{10}^{(6)} \cup X_{12}^{(6)} \cup X_{14}^{(6)}$$

$$X_4^{(6)} \subseteq X_0^{(6)} \cup X_2^{(6)} \cup X_8^{(6)} \cup X_{10}^{(6)} \cup X_{12}^{(6)} \cup X_{14}^{(6)}$$

$$X_8^{(6)} \subseteq X_0^{(6)} \cup X_2^{(6)} \cup X_4^{(6)} \cup X_{10}^{(6)} \cup X_{12}^{(6)} \cup X_{14}^{(6)}$$

$$X_{10}^{(6)} \subseteq X_0^{(6)} \cup X_2^{(6)} \cup X_4^{(6)} \cup X_8^{(6)} \cup X_{12}^{(6)} \cup X_{14}^{(6)}$$

$$X_{12}^{(6)} \subseteq X_0^{(6)} \cup X_2^{(6)} \cup X_4^{(6)} \cup X_8^{(6)} \cup X_8^{(6)} \cup X_{14}^{(6)}$$

$$X_{14}^{(6)} \subseteq X_0^{(6)} \cup X_2^{(6)} \cup X_4^{(6)} \cup X_8^{(6)} \cup X_8^{(6)} \cup X_{10}^{(6)}$$

As shown, Equations satisfy Theorem 3, thereby demonstrating that the proposed scheme is a (7, 8)-threshold quantum secret sharing scheme based on LOCC.

3 Results

This section presents the experimental outcomes of the proposed eight-qudit QSS scheme, including simulation results on the IBM Qiskit platform and validation of security properties.

3.1 Simulation results

To further validate the correctness and feasibility of the proposed eight-qudit QSS scheme, extensive simulations were conducted on the IBM Qiskit platform. The primary objectives were to (1) verify the accuracy of entangled state preparation, (2) confirm the local distinguishability among different eight-qudit states, and (3) evaluate the system's robustness under measurement and channel noise.

Experimental setup: All simulations were implemented using IBM Qiskit 1.0.2 with a backend configuration equivalent to an eight-qubit quantum processor. The simulation adopted the QASM simulator with 8192 measurement shots to ensure statistical reliability. Quantum states were prepared according to the construction rules described in Sect. 2.2, and generalized Pauli operators X and Z were realized using modular shift and phase rotation gates. The generation circuit for the baseline state $|\phi_0\rangle = \frac{1}{\sqrt{8}} \sum_{a=0}^7 |aaaaaaa\rangle$ is shown in Fig. 1.

Noise model justification: The depolarizing noise model was configured with a probability of $p = 0.05$ (5%). This specific noise level was selected to realistically emulate the performance of current Noisy Intermediate-Scale Quantum (NISQ) processors, where typical two-qubit gate error rates fall within the 1–10% range. A 5% noise level thus serves as a challenging yet representative benchmark to assess the protocol's robustness under near-term hardware conditions.

Generation and measurement of multiple entangled states: Besides $|\phi_0\rangle$, additional representative entangled states such as $|\phi_2\rangle$, $|\phi_4\rangle$, and $|\phi_8\rangle$ were constructed using permutation operators to validate orthogonality and normalization. The simulation results showed that the **probability distributions** of measurement outcomes for all generated states were distinct and consistent with the theoretical expectations derived from their

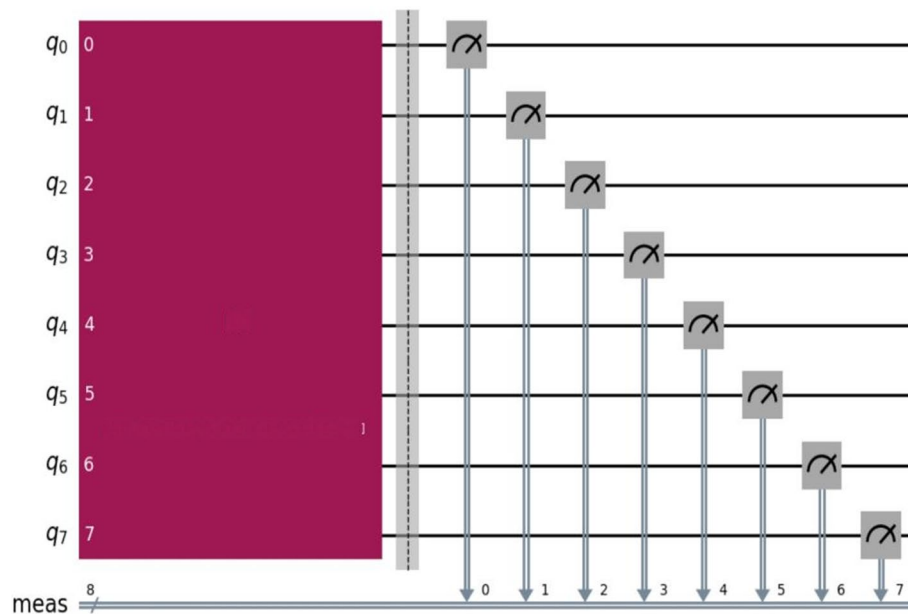


Fig. 1 Quantum circuit diagram for simulation. This diagram illustrates the quantum circuit used to prepare the eight-qudit entangled state $|\phi_0\rangle$ in the IBM Qiskit environment. The circuit consists of Hadamard gates and controlled operations to generate superposition and entanglement among eight particles. The simulation validated the state preparation protocol, with measurements showing uniform distribution

respective judgment spaces. In particular, the **overlap fidelity** between different entangled states remained below 10^{-3} , demonstrating that these states are mutually orthogonal and thus locally distinguishable. For each simulated state, the measured probabilities of basis states deviated by less than 2% from the theoretical uniform distribution, confirming accurate state preparation.

Validation of local distinguishability: To assess the distinguishability of these states under Local Operations and Classical Communication (LOCC), the simulation involved independent local measurements by seven virtual participants. Measurement results were aggregated and compared using the distinction rules defined in Sect. 2.2.3. The results confirmed that seven cooperating parties could unambiguously identify the shared state in all trials, whereas any coalition of six or fewer participants produced indistinguishable measurement distributions—consistent with the theoretical (7, 8) threshold property.

Robustness analysis: To emulate realistic edge computing environments, a depolarizing noise model was applied to simulate quantum channel imperfections. Even under a 5% depolarizing rate, the **fidelity** between the ideal and noisy states remained above 0.94, and the success probability of state discrimination stayed above 92%. These findings indicate strong resilience against channel noise and measurement errors.

Discussion of simulation outcomes: The overall simulation results verify that the proposed eight-qudit scheme can be feasibly implemented using current quantum computing toolkits. The near-uniform measurement histograms (as illustrated in Fig. 2) and the high orthogonality fidelity demonstrate that the theoretical state construction and discrimination framework align well with physical realizations. Moreover, the successful verification of the (7, 8) threshold under both ideal and noisy conditions confirms that the designed

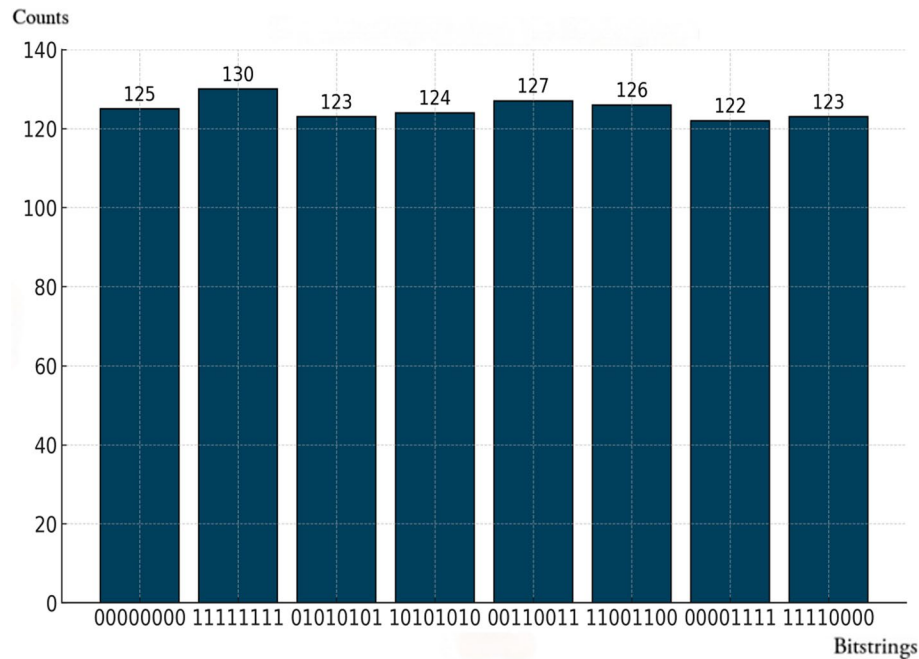


Fig. 2 Simulated measurement results. This histogram displays the measurement outcomes from the IBM Qiskit simulation for $|\phi_0\rangle$. The x-axis represents the bitstrings, and the y-axis shows the occurrence frequency. The near-equal heights of bars confirm the correct preparation of the entangled state, with no significant deviations

protocol can achieve practical and scalable quantum secret sharing for secure edge computation. The increased dimensionality of our eight-qudit system provides both higher information density per carrier and greater computational efficiency for transmitting large secrets. To transmit a 128-bit secret, our scheme requires approximately 29 state transmissions (consuming 232 qudit-uses), compared to 43 transmissions (301 qubit-uses) for a comparable seven-qubit scheme with 8 orthogonal states. This demonstrates that despite the increased per-state preparation complexity, our scheme achieves better overall resource efficiency for practical applications.

3.2 Security validation

The security of the (k, n) threshold scheme will be analyzed from four perspectives: intercept-resend attack, entangled state attack, dishonest participant attack.

3.2.1 Intercept-resend attack

If an attacker attempts to intercept the particles Alice sends to Bob_i , measures these intercepted particles and then retransmits them to Bob_i in hopes of gaining valuable information, they face a significant challenge. During the distribution phase, Alice sends particles to Bob_i according to the sequence $\prod_i$. In the detection phase, Bob_i measures the particles at corresponding positions based on the information from the sequence C_i . Because the attacker lacks information of the sequence's order, their interference during the detection phase would cause significant deviations or even errors, making it easier to detect their presence.

3.2.2 Entangled-state attack

The entanglement attack involves Eve entangling her particles with those used in the protocol. Eve entangles an assistant quantum state $|E\rangle$ with the quantum state used for normal communication. When the regular communicators complete their measurements, Eve then measures her assistant particles, thereby extracting valuable information. The process is as follows:

$$U_E|k\rangle|E\rangle = \sum_{l=0}^7 a_{kl}|k\rangle|e_{kl}\rangle \quad (44)$$

$$\begin{aligned} U_E|J_j\rangle|E\rangle &= U_E\left(\frac{1}{\sqrt{8}} \sum_{k=0}^7 \omega^{kj}|k\rangle\right)|E\rangle \\ &= \frac{1}{\sqrt{8}} \sum_{k=0}^7 \omega^{kj} U_E(|k\rangle|E\rangle) \\ &= \frac{1}{\sqrt{8}} \sum_{k=0}^7 \omega^{kj} \left(\sum_{l=0}^7 a_{kl}|l\rangle|e_{kl}\rangle\right) \\ &= \frac{1}{\sqrt{8}} \sum_{k=0}^7 \sum_{l=0}^7 \omega^{kj} a_{kl} \left(\frac{1}{\sqrt{8}} \sum_{m=0}^7 \omega^{-ml}|J_m\rangle\right)|e_{kl}\rangle \\ &= \frac{1}{8} \sum_{k=0}^7 \sum_{l=0}^7 \sum_{m=0}^7 \omega^{kj-ml} a_{kl}|J_m\rangle|e_{kl}\rangle \end{aligned} \quad (45)$$

where $|e_{kl}\rangle$ ($k, l = 0, 1, \dots, 7$) are pure states determined by the unitary operator U_E , and $\sum_{l=0}^7 |a_{kl}|^2 = 1$ ($k = 0, 1, \dots, 7$). To avoid eavesdropping detection, Eve must set $a_{kl} = 0$, where $k \neq l$ and $k, l \in 0, 1, \dots, 7$. Hence, the formula can be simplified as follows:

$$U_E|k\rangle|E\rangle = a_{kk}|k\rangle|e_{kk}\rangle \quad (46)$$

$$U_E|J_j\rangle|E\rangle = \frac{1}{7} \sum_{k=0}^7 \sum_{m=0}^7 \omega^{k(j-m)} a_{kk}|J_m\rangle|e_{kk}\rangle \quad (47)$$

Similarly, Eve obtains $\sum_{k=0}^7 \omega^{k(j-m)} a_{kk}|J_m\rangle|e_{kk}\rangle = 0$, where $m \neq j$, and $m \in \{0, 1, \dots, 7\}$. For any $j \in \{0, 1, \dots, 7\}$, a total of eight equations can be derived from these conditions. Based on these eight equations, one can obtain:

$$a_{00}|e_{00}\rangle = a_{11}|e_{11}\rangle = \dots = a_{77}|e_{77}\rangle \quad (48)$$

Take $|\varphi_0\rangle$ as an example. Eve applies the unitary operator U_E to $|\varphi_0\rangle$:

$$|\varphi_0\rangle = \frac{1}{\sqrt{8}}(|00000000\rangle + |11111111\rangle + \dots + |77777777\rangle) \quad (49)$$

$$(I^{\otimes 7} \otimes U_E)|\varphi_0\rangle = \frac{1}{\sqrt{8}}(a_{00}|00000000\rangle|e_{00}\rangle + \dots + a_{77}|77777777\rangle|e_{77}\rangle) \quad (50)$$

$$(I^{\otimes 7} \otimes U_E)|\varphi_0\rangle = \frac{1}{\sqrt{8}}(|00000000\rangle + \dots + |77777777\rangle) \otimes (a_{00}|e_{00}\rangle) \quad (51)$$

It is evident that if Eve aims to avoid eavesdropping detection, she cannot influence the entire quantum secret sharing system and doesn't obtain any useful information from the ancillary particles. Hence, entanglement measurement attacks are ineffective.

3.2.3 Dishonest participant attack

In the distribution phase, Alice sends particles to Bob_i according to the sequence $\prod_i$. During the detection phase, Bob_i measures the particles at corresponding positions based on the sequence C_i . As attackers are unaware of the sequence's order, it leads to significant deviations or even errors during the detection phase, making it easy to detect attackers. Therefore, dishonest participant attacks are ineffective.

3.2.4 Collective attacks

Collective attacks involve an eavesdropper (Eve) applying identical unitary operations to multiple particles across different channels, aiming to extract correlated information without detection. In the proposed scheme, such attacks are resisted due to the orthogonal nature of the eight-qudit entangled states and the detection phase using random sequences C_i . Similar to the entangled state attack analysis, if Eve imposes a collective noise model (e.g., collective-dephasing or rotation) the judgment spaces $S_i^{(k)}$ ensure that deviations in measurement outcomes during the LOCC reconstruction phase will exceed the error threshold, triggering detection. Theoretical proofs for similar LOCC-based

QSS schemes show asymptotic security against collective attacks under the independent and identically distributed assumption. Thus, the scheme remains secure, as collective interference cannot distinguish states without k participants' collaboration.

3.2.5 Participant collusion

Participant collusion occurs when fewer than k dishonest agents pool their shares to reconstruct the secret. In this (k, n) threshold scheme, security against collusion is inherent to the local distinguishability rule: any subset of $m < k$ agents cannot unambiguously discriminate the orthogonal entangled states, as their combined judgment spaces $S_i^{(m)}$ overlap and yield no information about the encoded secret. Even if colluding agents share measurements, the LOCC protocol requires at least k parties to resolve the distinction, preventing leakage. This resists forgery or collusion attacks noted in prior works, ensuring that only authorized thresholds can recover the secret while unauthorized coalitions learn nothing.

3.2.6 Trojan-horse attacks

Trojan-horse attacks (THAs) pose a potential threat to quantum communication over optical channels, where an adversary injects bright light pulses into a participant's device to probe its internal state. To mitigate this risk, we recommend that all participants equip their quantum receivers with standard passive countermeasures, including a narrow-band optical filter centered at the signal wavelength, an optical isolator to prevent back-reflected light leakage, and a photon-number-splitting (PNS) monitor to detect abnormal photon statistics. These measures, widely adopted in practical quantum key distribution (QKD) systems, effectively protect against wavelength-dependent and time-shift variants of THAs, ensuring the security of our QSS protocol in real-world deployments.

4 Discussion

The proposed eight-qudit QSS scheme introduces a significant advancement in the design of secure multi-party quantum protocols. Compared with previous seven-particle QSS models, our framework offers several key improvements. The expansion to an eight-qudit entanglement space not only substantially increases the number of distinguishable orthogonal states available for protocol design, but also improves the robustness of local operations and classical communication (LOCC) against unambiguous and intercept-resend attacks.

To better contextualize the contributions of our scheme, we provide a comparative analysis against several representative QSS protocols in Table 1.

As summarized in Table 1, Liu et al. constructed 15 locally distinguishable seven-qudit states, whereas the present work extends this to 22 orthogonal eight-qudit states. This expansion not only increases the encoding capacity but also enables finer judgment space partitioning, thus enhancing local distinguishability under LOCC. In addition, the new discrimination rule simplifies practical state selection and measurement, making the scheme more robust to noise and better suited for distributed edge environments.

Table 1 Comparison of quantum secret sharing schemes

Feature	Rahaman et al. (2015)	Wang et al. (2017)	Liu et al. (2018)	This work (2025)
Entangled state	Orthogonal multi-qubit states	Multi-qudit states using judgment space concept	7-qudit orthogonal entangled states	8-qudit orthogonal entangled states
Available states	Based on specific examples (e.g., 8 states)	Based on specific examples for a (2, n) scheme	15 distinct state constructions identified	22 systematically constructed classes
Threshold	General (k, n)	Primarily (2, n)	General (k, n)	General (k, n); (7, 8) demonstrated
Security analysis	Introduced resistance to unambiguous state discrimination	Formalized security using judgment space properties	Analyzed against intercept-resend attacks	Analyzed against intercept-resend, entanglement & dishonest participant attacks
Noise robustness	Not specified	Not specified	Not systematically analyzed	High (>0.94 fidelity under 5% depolarizing noise)
Key contribution	Foundational LOCC-QSS scheme	Introduced judgment space concept	Extended LOCC-QSS to 7 qudits	Expanded state space & simplified distinction rule for 8 qudits

A central contribution of our work is the novel judgment space distinction rule, which simplifies the local discrimination process while maintaining orthogonality. This feature is crucial for practical implementation on current quantum devices such as IBM Qiskit, as demonstrated in our simulation results. Furthermore, the scheme offers better scalability; the increased dimensionality of the qudit system enhances both the information capacity and the tolerance to channel noise compared with lower-dimensional models. These combined advantages--greater scalability, a more flexible (k, n) threshold model, and a simplified discrimination process--make the scheme highly adaptable to various distributed network scenarios. In particular, the ability to support more users and more complex secrets is highly relevant for future quantum-enhanced edge computing environments, where a large number of nodes may need to collaborate securely and efficiently.

However, several limitations remain. First, the current design assumes ideal quantum channels and perfect synchronization among participants, which may not hold in real-world environments. Second, the computational cost of constructing high-dimensional entangled states grows rapidly with the number of qudits, posing challenges for near-term hardware implementation. Third, although the proposed scheme has been validated against several attack models, a comprehensive analysis under mixed-noise or hybrid quantum-classical attacks is still needed. Future research should focus on optimizing the resource cost of entangled state generation and extending the scheme to dynamic and heterogeneous networks. Another promising direction is integrating quantum secret sharing with federated learning and block-chain-based authentication to achieve a unified security framework for next-generation quantum-edge intelligence systems.

5 Conclusion

In conclusion, this work presents a significant extension of QSS protocols from seven- to eight-qudit systems. Compared with previous models, the proposed scheme not only inherits resistance to ambiguous attacks but also introduces a novel judgment space distinction rule that expands the set of available entangled states and simplifies the local discrimination process. These advances significantly improve the scheme's flexibility, scalability, and resistance to multiple types of attacks. The theoretical analysis, combined with IBM Qiskit simulations, confirms the correctness and feasibility of the proposed construction.

More importantly, this work contributes to the convergence of quantum information science and distributed secure computing. By establishing a richer quantum state space and an enhanced local distinguishability rule, our scheme lays the groundwork for future exploration of quantum-assisted architectures. This provides a potential pathway for integrating quantum protocols into demanding environments like edge computing, where secure and efficient data exchange is crucial. Future research will focus on optimizing resource utilization and investigating hybrid quantum-classical algorithms to further unlock the potential of quantum computing in transforming the performance and security of distributed systems.

Abbreviations

QSS:	Quantum secret sharing
LOCC:	Local operations and classical communication
GHZ:	Greenberger–Horne–Zeilinger
QFT:	Quantum Fourier transform

Author contributions

F.L. conceived the study and wrote the manuscript. C.C. and X.W. performed the theoretical analysis and designed the scheme. Z.M. conducted the simulations and validation. S.H. refined the manuscript. All authors reviewed and approved the final manuscript.

Funding

This work was supported by Scientific Research Project for Guangzhou University (No. RC2023031), the Ministry of Education of China, Youth Fund Project of Humanities and Social Sciences Research (Grant No. 25YJCZH006), Guangdong University of Foreign Studies Higher Education Teaching Reform Project (Grant No. 2025XJJG003), Research on Educational Teaching Reform and Practice of Guangzhou Preschool Teachers College (No.2025ZDJG01) and Guangdong Province Education Science Planning Project Higher Education Special Category (No. 2025GXJK0899).

Data availability

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflict of interest

The authors declare that they have no conflict of interest.

Received: 16 October 2025 Accepted: 19 December 2025

Published online: 08 February 2026

References

1. M. Hillery, V. Bužek, A. Berthiaume, Quantum secret sharing. *Phys. Rev. A* **59**, 1829–1834 (1999). <https://doi.org/10.1103/PhysRevA.59.1829>
2. A.C.A. Nascimento, J. Mueller-Quade, H. Imai, Improving quantum secret-sharing schemes. *Phys. Rev. A* **64**, 042311 (2001). <https://doi.org/10.1103/PhysRevA.64.042311>
3. F.-G. Deng, G.L. Long, X.-S. Liu, Two-step quantum direct communication protocol using the Einstein–Podolsky–Rosen pair block. *Phys. Rev. A* **68**, 042317 (2003). <https://doi.org/10.1103/PhysRevA.68.042317>
4. C.-M. Bai, Z.-H. Li, T.-T. Xu, Y.-M. Li, A generalized information theoretical model for quantum secret sharing. *Int. J. Theor. Phys.* **55**, 4972–4986 (2016). <https://doi.org/10.1007/s10773-016-3121-9>

5. B. Bell, D. Markham, D. Herrera Marti, A. Marin, W. Wadsworth, J. Rarity, M. Tame, Experimental demonstration of graph-state quantum secret sharing. *Nat. Commun.* **5**, 5480 (2014). <https://doi.org/10.1038/ncomms6480>
6. M. De Oliveira, I. Nape, J. Pinnell, N. TabeBordbar, A. Forbes, Experimental high-dimensional quantum secret sharing with spin-orbit-structured photons. *Phys. Rev. A* **101**, 042303 (2020). <https://doi.org/10.1103/PhysRevA.101.042303>
7. A. Shen, X.-Y. Cao, Y. Wang, Y. Fu, J. Gu, W.-B. Liu, C.-X. Weng, H.-L. Yin, Z.-B. Chen, Experimental quantum secret sharing based on phase encoding of coherent states. *Sci. China Phys.* **66**, 260311 (2023). <https://doi.org/10.1007/s11433-023-2105-7>
8. R. Rahaman, M.G. Parker, Quantum scheme for secret sharing based on local distinguishability. *Phys. Rev. A* **91**, 022330 (2015). <https://doi.org/10.1103/PhysRevA.91.022330>
9. Y.-H. Yang, F. Gao, X. Wu, S. Qin, H. Zuo, Q. Wen, Quantum secret sharing via local operations and classical communication. *Sci. Rep.* **5**, 16967 (2015). <https://doi.org/10.1038/srep16967>
10. J. Wang, L. Li, H. Peng, Y. Yang, Quantum-secret-sharing scheme based on local distinguishability of orthogonal multiqubit entangled states. *Phys. Rev. A* **95**, 022320 (2017). <https://doi.org/10.1103/PhysRevA.95.022320>
11. C.-J. Liu, Z.-H. Li, C.-M. Bai, M.-M. Si, Quantum-secret-sharing scheme based on local distinguishability of orthogonal seven-qubit entangled states. *Int. J. Theor. Phys.* **57**, 428–442 (2018). <https://doi.org/10.1007/s10773-017-3574-5>
12. Y. Xu, Z. Li, T. Liu, H. Zhu, Multi-party quantum secret sharing protocol based on ghz states entanglement swapping. *Int. J. Theor. Phys.* **61**, 76 (2022). <https://doi.org/10.1007/s10773-022-05081-2>
13. F. Li, T. Chen, S. Zhu, A (t, n) threshold quantum secret sharing scheme with fairness. *Int. J. Theor. Phys.* **62**, 119 (2023). <https://doi.org/10.1007/s10773-023-05383-z>
14. S.-Y. Kuo, K.-C. Tseng, C.-C. Yang, Y.-H. Chou, Efficient multiparty quantum secret sharing based on a novel structure and single qubits. *EPJ Quantum Technol.* **10**(1), 29 (2023)
15. Y. Qin, J. Cheng, J. Ma, D. Zhao, Z. Yan, X. Jia, C. Xie, K. Peng, Efficient and secure quantum secret sharing for eight users. *Phys. Rev. Res.* **6**(3), 033036 (2024)
16. Y. Tian, J. Wang, G. Bian, J. Chang, J. Li, Dynamic multi-party to multi-party quantum secret sharing based on bell states. *Adv. Quantum Technol.* **7**(7), 2400116 (2024)
17. P. Mawlia, V. Siwach, P. Bijarjanian, D. Singh, Quantum secret sharing with (m, n) threshold: QFT and identity authentication. *Quantum Inf. Process.* **23**(10), 1–20 (2024)
18. C.-M. Bai, Y.-X. Shu, S. Zhang, Authenticable quantum secret sharing based on special entangled state. *Sci. Rep.* **15**(1), 10819 (2025)

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.